

Multimodal AI Model Exploration & Web Application Development

In this assignment, you have to explore different AI model providers and different input-output modalities. The goal is to understand how modern AI models work across text, image, audio, and video-based tasks.

AI Platforms You Can Use

- OpenAI
- Claude
- Google Gemini
- Groq
- OpenRouter
- Hugging Face

Supported Modalities

- Text → Text
- Text → Image
- Image → Text
- Text → Audio
- Audio → Text
- Text → Video
- Video → Text

Optional Modalities (Not Required):

- Image → Image
- Audio → Audio

Task 1: Create a New Notebook

Prepare a fresh notebook and choose different models that were not used in the live class. For each selected modality:

- Select a suitable model.
- Load or call the model using API/local inference.
- Provide sample input.
- Generate output successfully.
- Show the final result clearly in the notebook.

Task 2: Build a Web Application

After successfully testing models in the notebook, create a web application where users can provide inputs in different modalities.

Your web app should support:

- Text input
- Image upload
- Audio upload/recording
- Video upload

Possible Outputs:

- Text response
- Generated image
- Generated audio
- Transcribed text
- Generated video
- Video summary/description

Expected Final Submission

- Jupyter/Colab notebook
- Web application code
- README file
- Screenshots or demo video
- List of models used
- Explanation of each modality

Evaluation Criteria

- Correct model selection
- Successful notebook execution
- Multiple modality coverage
- Web app functionality
- Code quality
- README documentation
- Creativity and presentation

Final Goal

By the end of this assignment, you should be able to build a working multimodal AI system that can accept text, image, audio, or video input and generate meaningful AI-powered outputs.